

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458400.8632 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.063
Transit depth (Rp/Rs)^2	4.58 +/- 2.68 [%]
Orbital Inclination (inc)	85.1 +/- 2.67
Airmass coefficient 1 (a1)	1.34 +/- 0.82
Airmass coefficient 2 (a2)	-0.12 +/- 0.12
Scatter in the residuals of the lightcurve fit is	67.04 %
Best Comparison Star	#3 - [151, 38]
Optimal Aperture	1.58
Optimal Annulus	6.25
Transit Duration (day)	0.072 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.063 vs 0.1646 (deviation 0.39σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.08σ)
Mid-time O–C	10.9 min
Depth SNR	1.7
Residual scatter	67.04 %

Light curve

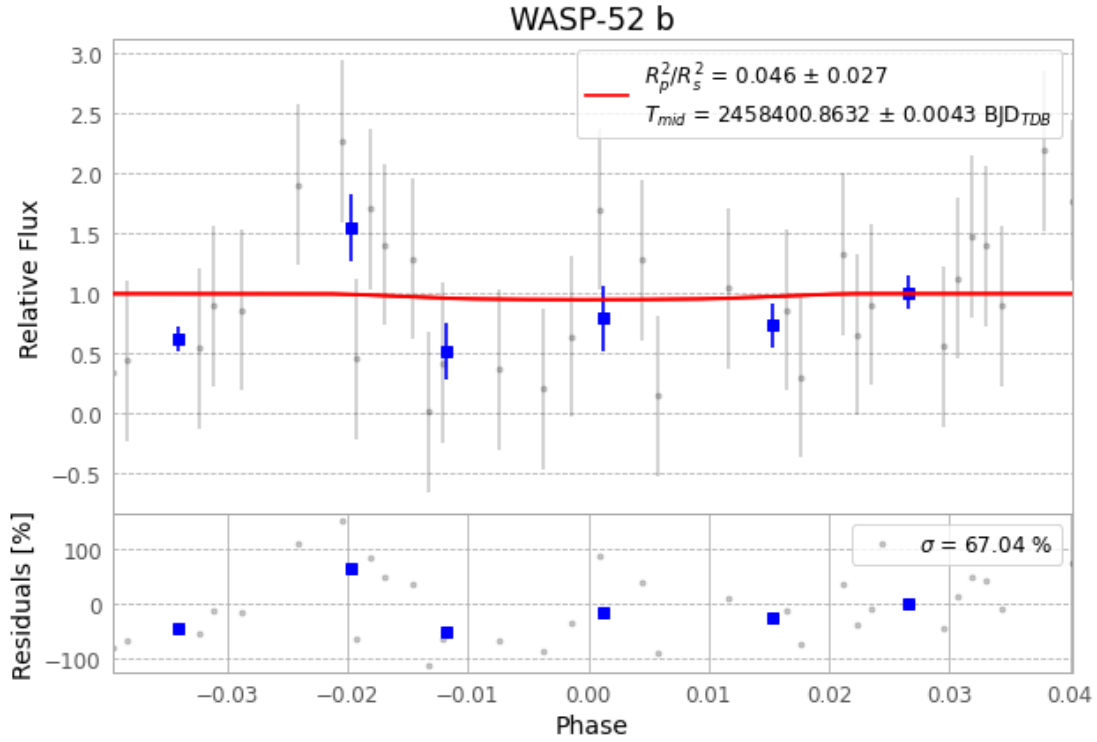


Figure 1 — FinalLightCurve_WASP-52 b_2018-10-08.png (SHA-256 e30f66112782eb92...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [619, 436], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 77ec3e90f8f7a0af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

74 FITS frames (49 MB), WASP-52181009064512.FITS → WASP-52181009102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181009.log (4f304ef50648...), FinalLightCurve_WASP-52 b_2018-10-08.png (e30f66112782...), FinalLightCurve_WASP-52 b_2018-10-08.csv (67864bceb81e...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 18:04Z.